

Decoding the Invisible Forces of Electricity

A visual guide to understanding Ohm's Law and basic circuit dynamics.

Three Forces Drive Every Electrical System

V

Voltage provides the underlying push. It is the initial force that makes electrical movement possible.

I

Current is the actual movement. It represents the volume of electrical charge flowing through the system.

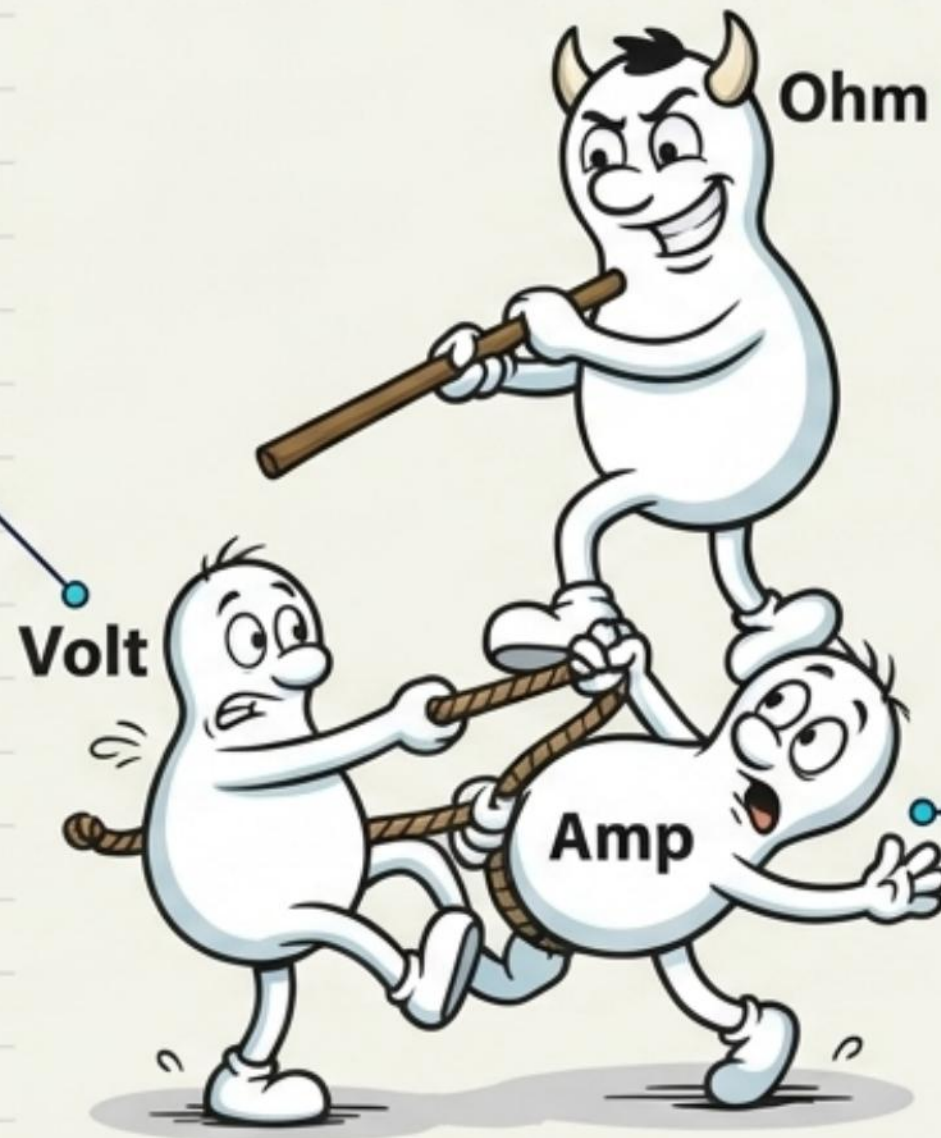
R

Resistance is the deliberate bottleneck. It restricts and controls the flow to do useful work.

Visualizing the Push, the Flow, and the Squeeze

The Electromotive Force.

Volt uses sheer effort to push the charge forward.



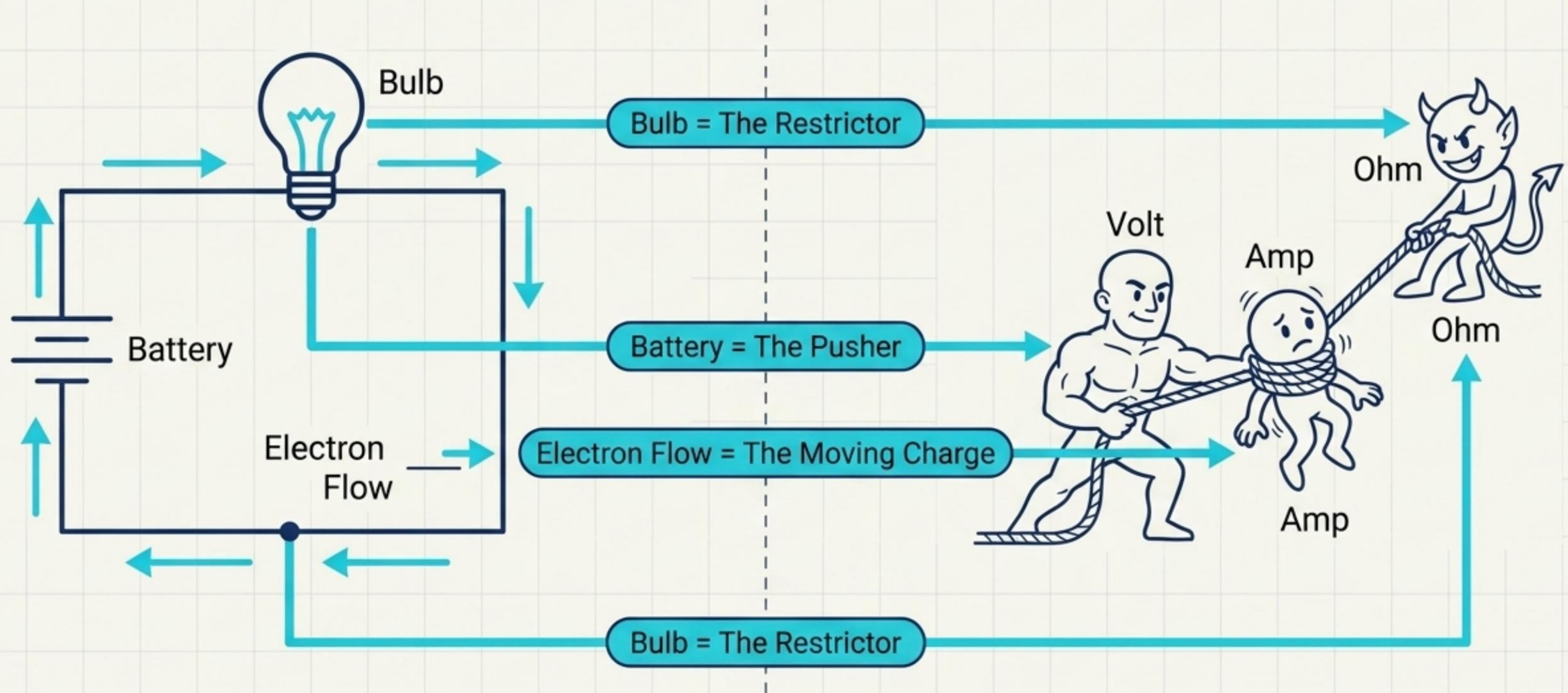
The Flow Restrictor.

By pulling the rope, Ohm creates a physical chokepoint that dictates exactly how much of Amp can pass through.

The Rate of Flow.

Amp is the actual charge traveling through the pathway, feeling the pressure.

Translating the Metaphor into a Physical Circuit



The Electricity Trinity Blueprint

	Voltage (V)	Current (I)	Resistance (R)
Definition	The pressure from an electrical circuit's power source.	The rate at which electrons flow past a point.	A material's opposition to the flow of electric current.
Analogy Role	The Push	The Flow	The Restrictor
Circuit Component	Battery	Flow of Electron	Bulb
Unit of Measure	Volts	Amps	Ohms

Ohm's Law Defines the Delicate Balance

If Volt pushes with more strength (Voltage increases), more of Amp gets shoved through the gap.

The flow (Current) increases.

$$V = I \times R$$

If Ohm pulls the rope tighter (Resistance increases), Amp gets squeezed harder.

The flow (Current) decreases.

Ohm's Law proves that the flow of electricity is always a direct negotiation between the strength of the push and the tightness of the squeeze.

Master the Circuit



$$V = I \times R$$



- Voltage initiates the push.
- Resistance determines the bottleneck.
- Current is the resulting flow.
- Everything is connected. Change one, and the entire system responds.